

BACHELORS OF COMPUTER APPLICATION

EXAM & LIBRARY Portal

Smart Competitive Exam Preparation & Digital Library Management
System

A premium full-stack offline web application built with HTML, CSS, JavaScript, Node.js, Express.js, and SQLite – unifying detailed exam prep, digital library workflows, and analytics into one seamless platform.

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Department **Bachelors of Computer
Application**

Platform Type **Offline Full-Stack
Ecosystem**

Project Overview & Objectives

01

⚠️ Current Problems

Students preparing for competitive exams lack a centralized, offline-capable platform for study assets.

❌ Scattered study resources

❌ No offline access capacity

❌ Poor library booking track

02

✂️ Portal Solution

Our unified platform bridges this gap, integrating syllabus trackers and borrowing tools under one screen.

✅ Centralized portal interface

✅ Smart digital library tracking

✅ Comprehensive diagnostics

03

🎓 Student Success

A structured ecosystem designed to systematically elevate performance and track objectives.

★ Organized prep milestones

★ Speedier search timelines

★ Measurable study progression

System Modules



Library System

Efficiently borrow, return, search, and manage books with structured inventory controls.



Detailed Syllabus

Organized unit-wise competitive exam layout trackers with built-in checkbox dynamics.



Exam Information

Complete directories for gov/private competitive exams, featuring criteria, patterns, and dates.



Notes & Materials

Central hub to upload and download PDFs, notes, question papers, and essential formula sheets.



Doubt Forum

Peer-to-peer discussion forum with options to ask, reply, comment, and pin critical subject topics.



Analytics Tracker

Graphical student progress charts, exam preparation logs, and download/borrow visual analytics.



Alert Notifications

Automated system notification alerts for book deadlines, upcoming exams, and syllabus milestones.



Smart Search

Fast keyword filtering system across library catalogs, exams, subjects, and authors.

Technology Stack & Architecture

FRONTEND TECHNOLOGIES

 HTML5

 CSS3

 JavaScript

 Chart.js

 Font Awesome

BACKEND & DATABASE STACK

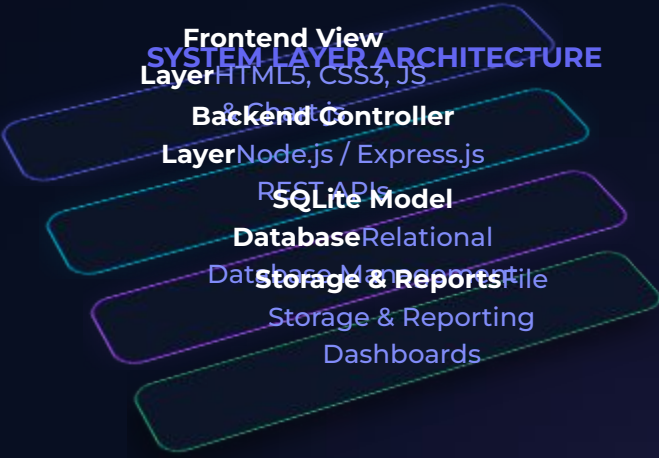
 Node.js Runtime

 Express.js

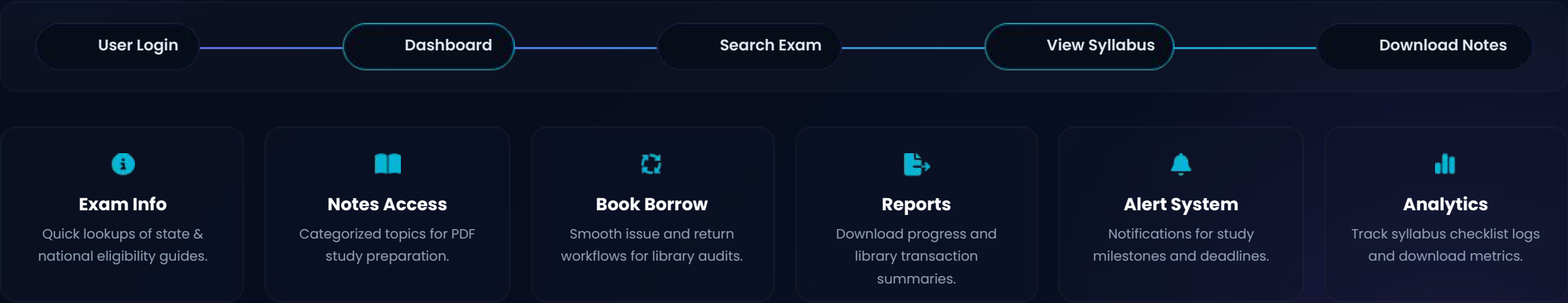
 SQLite (Offline DB)

 Multer Storage

 MVC Architecture



Features & User Workflow

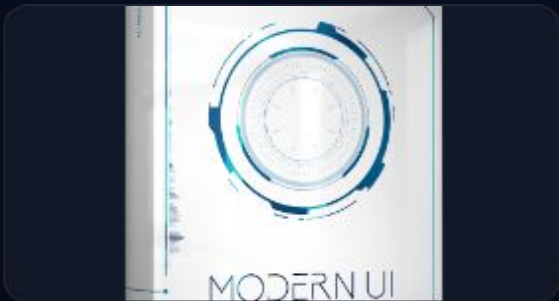


Dashboard & Interface Screenshots



Analytics Dashboard

Provides immediate diagnostic visual graphs mapping preparation progress and borrowing timeline histories.



Books Inventory

A modern search interface grid tracking available physical and digital study resources with search filters.



Smart Search System

Accelerates study processes through rapid, categorized index lookups across exams, subjects, and books.



Reports & Progress

Delivers printable syllabus records, download statistics, and book transaction logs to ensure high structural oversight.

Advantages & Future Roadmap

✓ Current Platform Advantages

- **Fully Unified Workspace** Centralizes separate resources into a single browser ecosystem.
- **Fast Offline Deployment** Runs entirely on local assets and networks via local SQLite instances.
- **Ultra-Responsive UI** Provides a fluid experience that scales from monitors to tablet layouts.

🚀 Future Expansion Roadmap

- **AI Study Planner** Incorporates localized algorithms for custom syllabus completion flows.
- **QR Integration System** Adds scanner support to match rapid check-out cards for physical libraries.
- **Dynamic Interactive Mock Exams** Integrates countdown quiz parameters directly inside student dashboards.

Project Conclusion

- 1 Unified Student Workspace** Combines standard catalog borrowing tools with active study syllabus progress metrics.
- 2 Robust Offline Architecture** Optimized performance using lightweight MVC patterns running on Express and SQLite databases.
- 3 Scalable Foundation Layout** Ready for cloud migration, customized machine learning modules, and native mobile packaging.



Thank

We welcome your valuable feedback and

You!



Presented By: **Abhishek Tiwari**



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